

Research Design 2026 — Session 2 teaching notes

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Questions, and where the data comes from. Tue 22 September 2026

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Session 2. Questions, and where the data comes from

Extended teaching notes

Companion to `questions.pdf` (50 slides) and `questions_notes.pdf` (slides + short presenter notes). This is the long version: what to say, what to ask, what answers to expect, and what to cut when you run out of time.

Slide numbers match the footer in `questions.pdf`. Slides labelled *Roadmap* are auto-generated section dividers; say the section name and move on. Slide 28 is a full-bleed image.

1. The shape of the evening

Unlike session 1, this one has a paper discussion, which means the slot is genuinely full.

	Block	Slides	Planned
18:00	Arrival, settling, the course map	1–2	15 min
18:15	A. From topic to question; theory, mechanisms, levels	3–25	55 min
19:10	Break		10 min
19:20	B. Where the data comes from	26–42	40 min
20:00	C. Paper: The Parable of Google Flu	43–47	45 min
20:45	Memo 2, next week, questions	48–50	15 min

That is 180 minutes exactly, with no slack. You will need the cut list.

Priority, if you have to choose

Block B is the one to protect. Memo 2 comes straight out of it, and it is the material that exists nowhere else in the programme. Block A is the most familiar material in the course and the easiest to compress; the paper discussion is scripted and can absorb five minutes of trimming without collapsing.

Cut in this order:

1. **“Evaluating theories” (12)**, 2 min. Three words on a slide; say them over the previous slide.
2. **“Stories, RQs, and theories” (10)**, 2 min. Folds into “Generating theories” without loss.
3. **“Why bother about theory?” (20)**, 3 min. The point is made on slide 19.
4. **The hypotheses 2×2 (16–17)**, 8 min. Genuinely good, and the most expensive thing in Block A. Cut only if you are properly behind at the break.
5. **Shorten the redesign step (46)** from 8 minutes to 5.

That is up to 20 minutes without touching Block B.

Before you walk in

- **Ask how many have read the paper**, honestly, at slide 44, and adapt. Nothing was handed in and nothing is marked, so this is real information rather than a rhetorical question. If it is under half, present more of the design yourself and shorten the prompts.
- Memo 1 came in Monday night. Skim it before class: what they chose to audit tells you what landed last week.
- **The two % TODO: figures** (CDC vs GFT on slide 38, the Xbox skew on slide 36) are still missing. The frames work without them. If you find them, they slot straight in.

2. Block A: from topic to question (slides 3–25)

2. Where we are

Ten seconds. The five-questions slide with question 2 highlighted. It is what makes the course feel like one argument rather than a list of topics.

Say what the shape of tonight is: two halves that look unrelated and are not. How to get to a question worth asking, and where the data that might answer it actually came from.

4. What is a research question?

The most useful twenty minutes in the course for students who have never written a design. Build the ladder one overlay at a time.

- **Topic > RQ > Theory** is the version they will have been taught.
- **Previous evidence > argument > RQ > hypotheses** is closer to how research works.
- **Previous > “anecdotal argument” > RQ > proper theory > hypotheses** is the honest one.

The third version is the one that matters, because it gives them permission to start from a hunch, which is where everyone actually starts. Saying so out loud removes a lot of anxiety about whether they are doing it “properly”.

Callback worth making: the hotel and Spotify examples from last week entered this ladder in the middle, at the data, and had to work backwards to the question.

5. The worked example (currently Charlie Kirk)

Marked % SWAPPABLE: in the source. The pedagogical point is the three-way split, not the topic.

Name what the sub-questions are, because this is last week's triad reappearing inside a single project:

- RQ1 (has political violence increased?) is **descriptive**
- RQ2 (more likely by individual young men?) is **descriptive/distributional**
- RQ3 (is online radicalization the cause?) is **causal**

Most students will present something like RQ3 as their question and then answer something like RQ1. That is exactly the mismatch you named in week 1, and here it is in a real example.

A question worth asking: which of the three is hardest? The room usually says RQ3. The honest answer is RQ1, and next week is about why. That inversion lands well and sets up session 3.

6–8. Good research questions

6 gives the two criteria, and both have to hold at once: answerable but irrelevant is trivia, relevant but unanswerable is an essay rather than a design.

7 has the three practical tests. The first is the most useful and the least practised: **imagine the result before you have it**. If every possible answer is uninteresting, the question is wrong, and you can know that tonight rather than in March.

Test 2 (feasibility) is where the second half of the session lands. “Do you have the data” is not a yes or no. It is: does it exist, can you get it, and does it mean what you need it to mean.

8 is the three-way contrast, and it is the best twenty seconds on the slide. Take the three bad versions one at a time and make the room name the defect before you do.

The third one deserves dwelling on. “*Why does online media drive political violence?*” presupposes that it does. **A question that assumes its own answer cannot be tested**, and students write these constantly, usually because the assumption feels obvious to them.

9. Generating RQs (exercise)

Pairs, three minutes, then four or five contributions. With forty people an open question dies; a pair turn does not.

The argument on the slide: how well students do depends more on their peers than on teaching quality.

What to listen for. Most first attempts compare good schools with bad schools, which confounds peers and teaching completely, since good schools tend to have both. The useful move is finding variation that separates the two. That is a preview of weeks 4 and 5, and worth flagging as such.

A good answer names the comparison, not just the topic. Hold them to that; it is the standard for the whole course.

10–12. Theories

10: the abstraction ladder. Students tend to think theory is something you either have or do not; it is something you climb towards from a story.

11: the fourth bullet is the practical one and worth repeating: *if you cannot tell a story out of the theory, you are not there yet*. It is a good diagnostic to reuse in the workshop.

Take the closing question (“how to identify a good theory?”) as a real question before turning.

12: three criteria. The third does the work in this course. **A theory that cannot fail cannot be tested, and a design that cannot fail is not a design.**

13–14. The Erasmus example

13 is a descriptive fact: Erasmus students do better in the labour market. Let them look, and ask what they would conclude.

14 walks the whole ladder, and **the two candidate theories are the point:** the same descriptive fact supports a “no effect, it is all household income” story and a “real effect, but only for high-income students” story.

Those are different theories with different observable implications, which is what makes them testable. **Ask what you would have to see to tell them apart.** That question is the entire second half of the course in miniature.

15–17. Hypotheses

15: the second sub-bullet is the half students skip. What would have to be true for you to abandon the theory? Anyone who cannot answer that has a hope, not a hypothesis. This returns in week 5 with pre-registration.

16: make them fill in the four cells before you reveal. Pairs, one minute. The theory is “cold **and** running”, so only one cell should show pain. **The point is that a theory implies a whole pattern, not a single observation.**

17: now the data disagrees, and informatively: pain on cold days regardless of running, no pain running on hot days. Running looks irrelevant; cold does not.

Two things to draw out. The theory was wrong in a **specific** way, which tells you what to think next. And you could only learn that because the theory made a prediction for all four cells: a vaguer theory would have survived.

(This pair is the most expensive block in Block A at about 8 minutes. It is also one of the best. Cut it only if you are properly behind.)

19–20. Computational methods and theory

19 answers the obvious objection to everything so far: why start with questions at all when you could let the data speak?

Answer it at design level rather than as a complaint about machine learning. A model predicting ice cream sales from drowning rates works fine and tells you nothing you can act on. Prediction without a why is genuinely useful (Blumenstock, last week) but it answers a different question, and confusing the two is the week 1 mismatch again.

This matters here because a good share of this cohort arrived wanting to do exactly the thing this slide qualifies. Do not sneer at it; locate it.

20 is conflict prediction: well resourced, plenty of data, performance well below the enthusiasm. The lesson is not that prediction is bad. It is that rare events, changing processes, and outcomes that get intervened on are hard to predict, and no amount of data fixes a problem that lives in the data-generating process. Week 5 closes on the same theme.

21–22. Mechanisms

21: mechanisms matter here for a practical reason. They generate sub-questions, and sub-questions are usually more answerable than the main one. If you cannot get at “does X cause Y”, you can often get at one step of the chain, and that is a real contribution rather than a retreat.

22: pairs, three minutes. The mechanism in the Erasmus case is informal networks.

Sub-questions that follow: do high-income Erasmus students end up in different sectors? do they report finding work through personal contacts more often? does the effect disappear where hiring is anonymised?

That last one is a design, not just a question. Point out that they have just done the thing week 5 is about.

23–24. Levels of explanation

Moved here from the measurement material, because levels belong with theory-building.

23: Leipzig, 1989. Ask why mass protests emerge and let it sit. Answers will mix levels freely: grievances, opportunity, someone being first, the state hesitating. That mixing is the material for the next slide.

24: sort them.

- **Macro:** economic decline, regime weakness, diffusion from Poland and Hungary.
- **Micro:** an individual deciding to turn up, given what they expect others to do.

The closing question is the good one. Macro explanations are not micro explanations in disguise; they can be about aggregate patterns no individual produces alone. But **a macro claim with no plausible micro story underneath is usually a correlation with ambition.**

Connect back to the unit of analysis from week 1: choosing a level is choosing a question.

25. Recap, then break

A genuine pause. The second half is a different topic and it is worth arriving there clean.

3. Block B: where the data comes from (slides 26–42)

This is the CSS core of the course and the part that exists nowhere else in the programme. Protect it.

27. Last week, in one line

Thirty seconds of recall. Designed data answers a question you had; found data answers a question someone else had, or none. Everything in this half follows from the right-hand column.

28. Readymade and custommade

Salganik's framing, and the best available.

Duchamp's *Fountain* is a urinal he did not make, seen and repurposed. Michelangelo's *David* was made for the purpose. **Both are art. They involve completely different skills.**

Found data is a readymade. The creative work is not in producing it. It is in recognising what an object built for someone else's purpose can and cannot be used for. That is a real skill, and it is what this half of the session teaches.

29–30. Two kinds of data

29 (designed): a cross-national survey and a hand-coded conflict dataset.

The property that matters: **there is a codebook**, and it exists because someone had to decide in advance what counted. Those decisions are documented, arguable, and revisable. You can go and read why a given event is or is not in the data. Expensive, slow, small. That is the trade.

30 (found): credit card transactions and social media posts.

Neither has a codebook, because nobody was answering a research question when the data was produced. There is no document defining what counts as a purchase or a protest post, because nobody needed one. Big, fast, cheap, always available. That is the other side of the trade, and it is why everyone wants to use them.

31. The asymmetry that matters

The sentence that generalises to their whole MA:

With designed data you know what the numbers were meant to mean, because someone wrote it down. With found data you have to reconstruct it, from the purpose of whoever built the system.

Facebook did not build a friendship graph so you could measure social capital. Spotify did not tag genres so you could study cultural convergence. **The purpose is still in the data**, and if you do not reconstruct it you will mistake a fact about the system for a fact about the world.

This is exactly the Spotify question from last week: is this a fact about music, or a fact about Spotify? It is also Memo 2. Say it slowly.

33. Salganik's ten characteristics

A checklist they will use for the rest of the MA, with no statistical idea in it.

Three on the left, seven on the right. That ratio is the honest summary of working with found data. **Do not present the right-hand column as reasons not to use it.** Every one is manageable if you know it is there before you design, and fatal if you discover it in March.

Four get proper treatment tonight (non-representative, drifting, algorithmically confounded, inaccessible). Dirty gets a mention. Incomplete and sensitive point forward to weeks 3 and 5.

34. Big, always-on, non-reactive

The three genuine attractions, each with its qualification attached, because the qualification is where the design work is.

Big. The sub-bullet is the one to stress: more data narrows the confidence interval around a biased estimate. It buys precision, not validity. Callback to Challenger: no number of failures would have helped.

Always-on. The strongest and least appreciated. If you had wanted to study behaviour before and after some sudden event, no survey would have been in the field beforehand. Platform data always is. This is a genuine capability that survey research cannot match.

Non-reactive. Cuts both ways: no social desirability bias, and no informed consent either. The second half is the seed of the week 5 ethics session.

35–36. Non-representative, and the Xbox poll

35: two layers of selection. Who is on the platform is not who is in the population; who *posts* is not who is on the platform. Students routinely notice the first and miss the second. The people who post about politics on any platform are a tiny and unusual subset.

Then ask the closing question and take answers **before** turning. Rooms usually split into “yes, useless” and “no, big enough to be fine”. Both are wrong, which is what makes the next slide worth showing.

36: the Xbox poll. About 350,000 respondents, roughly 93% male, overwhelmingly young. Raw numbers gave a large Romney lead, which was obviously wrong. Adjusted using what was known about who was missing, it tracked the campaign closely and called the outcome.

The lesson is not that non-representative is fine. It is that **representativeness is a property of the design, not of the sample.** What made it work was knowing precisely *how* the sample differed, and having something external to anchor to.

Two consequences, cutting in opposite directions, and students usually hold only one:

- “*My Twitter data isn’t a random sample so it’s useless.*” Not necessarily, if you can characterise the skew.
- “*I have ten million observations so it doesn’t matter.*” It matters more. You will be very precisely wrong.

The adjustment itself is statistics and we are not doing it. **Our question is: do you know who is missing, and do you have anything to anchor to?** If not, the design fails at any sample size.

37–39. Drifting, and Google Flu

37: three sources of drift. The world changes, the people on the platform change, and **the platform changes**. The third is specific to found data and invisible from inside the dataset.

The closing line is the whole problem: a drifting measure does not throw an error. The numbers keep arriving, the series keeps updating, and the trend you are looking at is partly a trend in the instrument.

38: the Google Flu sequence. It worked from 2008, missed H1N1 in 2009 because that outbreak was not seasonal, then over-predicted almost every week from 2011, at more than double the CDC figure in the 2012–13 season.

Do not give away the diagnosis. The paper discussion is forty minutes away and this is its setup. The sequence matters: it worked, then failed specifically, then failed persistently. A system that had never worked would not be interesting.

If someone asks why, put it back to the room and hold it.

39: the two causes Lazer et al. name.

Big data hubris has the wider reach. The model was selected to correlate with CDC reports, not built to measure flu. Some winning search terms were seasonal rather than flu-related, which is fine until a flu season arrives out of season.

Algorithm dynamics is the CSS-specific one and the deeper problem. Google kept improving its product: suggested searches, autocomplete, health panels. Every improvement changed the behaviour being measured. **Nobody at Google was doing anything wrong, and the measure broke anyway.**

The transferable line, and the one to say slowly: with found data, **your instrument is owned and maintained by someone whose interests are not yours, and they will change it without telling you.**

Direct line to Spotify retagging genres last week. Say so explicitly.

40. Algorithmically confounded

Distinct from drift, and students merge the two. **Drift is the instrument changing over time. Algorithmic confounding is the instrument causing the thing you are measuring, even when nothing changes.**

Concrete: a friendship network on a platform that recommends friends partly reflects the recommender. Popularity measured where popular things are shown more will show more popularity. You cannot separate the behaviour from the system that elicited it by looking harder at the data.

Forward-reference DAGs in week 4 explicitly: **when you draw a diagram for a platform study, the platform is a node.**

If you have time, this is also the strongest argument for why an *experiment* on a platform (Guess et al., week 4) is worth so much more than observation on one.

41. Inaccessible, and the post-API age

Practical, and several of them will propose scraping something for the essay.

Two distinct problems. **Feasibility:** can you still get it in March? **Reproducibility:** if the data cannot be obtained again, nobody can check the finding, including you.

The design question for Memo 2 and for the essay: *can you actually get this, and could someone else get it again?* A project that fails this is much better abandoned in week 2 than in April.

Do not moralise. The honest position is that this is a real constraint on what computational social science can currently be, not a failure of nerve on the students' part.

42. Dirty, and sensitive

Two minutes, both flagged rather than covered.

Dirty: the point is not that data needs cleaning, which they know. It is that “we removed bots” is a choice with consequences that almost never appears in a paper, and different reasonable choices give different answers. That connects to researcher degrees of freedom in week 5.

Sensitive: flag it deliberately now, because essay proposals get written in week 4 and ethics is week 5. If someone plans to scrape and link personal data, they should meet the objection before they have invested in the design.

The credit card study from slide 30 is the callback: entirely legitimate research, and also a dataset that could identify almost anyone in it.

4. Block C: the paper (slides 43–47, 45 minutes)

44. Why this paper

One slide on why it is on the list: three pages, no equations, and it carries measurement validity, drift and algorithmic confounding in a single case, with a well-resourced team getting it wrong in public.

Then ask how many have read it, and mean it. Nothing was handed in and nothing is marked. If it is under half the room, present more of the design yourself and shorten the prompts rather than running a discussion that only six people can join.

45. Prompts, in pairs

Take them two at a time: four minutes in pairs, four or five contributions, then the next two. Open-floor does not work at this size.

1. **What was GFT actually measuring?** Not flu. Search behaviour correlated with CDC reports of flu. The gap between those two things is the whole paper.
2. **Whose behaviour changed?** Both, and that is the point. Google changed the product; users changed how they searched in response.
3. **Would more data have fixed it?** No. More of a mis-specified measure is more of a mis-specified measure. This is the “big” qualification from slide 34, and it is worth making them say it themselves.
4. **What would you have had to do in 2009 to see this coming?** The strongest answers: validate against something external and keep validating; treat the search engine as a changing instrument rather than a fixed one; publish the model so that failures are diagnosable.

46. Redesign it (groups of four)

The step that matters most. Asking students to *evaluate* a Science paper produces deference; asking what they would change produces engagement, and it is the same skill the workshop tests.

Good answers, roughly in order of sophistication:

- combine search data with CDC data rather than replacing it
- re-fit the model continuously instead of once
- choose search terms with a causal story rather than by correlation alone
- hold out whole seasons rather than weeks when validating
- publish the model so that others can diagnose failures

Take three groups, not twelve. Eight minutes, or five if you are behind.

47. The design lesson

Close on one sentence and hand to week 3.

Google Flu is usually told as a story about hubris. The more useful reading for this cohort is that it is a **measurement failure**: nobody ever established that the thing being counted was the thing they cared about, and no amount of data or engineering rescues that.

That is precisely question 3 of the course, which is next week.

5. Close (slides 48–50)

49. Before next week

The reading: Müller-Crepon et al. Be honest that it is a proper journal article rather than three pages, and tell them exactly what to read it for: **the concept, and how it gets turned into a measure**. They can skip the models entirely. Telling them what to skip makes it far more likely they read the rest.

Memo 2, in pairs, Aula Global by Monday night. This whole half-session applied to something they choose.

Push them towards data they could **genuinely obtain**. The feasibility question from slide 41 is the point, and a memo about a dataset they will never get is a wasted exercise.

- Good picks: a public API they have already used, an institutional dataset, municipal open data, something scraped from a site with sane terms.

- Bad picks: anything requiring platform access they do not have.

Remind them it has nothing to do with the final essay, and that pairs can change week to week.

6. After the session

Five minutes of notes is worth an hour of guessing later.

- **Where did you end up on the clock?** This session has no slack, so note which block overran.
- **How many had read the paper?** It is the first real test of an unenforced reading, and it decides whether the same format works for Müller-Crepon, which is a much longer piece.
- **Did the Xbox example land, or did it read as “so bias doesn’t matter”?** That misreading is the risk on slide 36, and if it happened you want to correct it in week 3 rather than in the essays.
- **What is in the Memo 1s?** They arrived Monday. Skimming them tells you whether the triad actually stuck.
- **Who spoke in the pair debriefs,** and whether the same voices are recurring from week 1.